

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Value at risk (VaR) is a statistic that measures and quantifies the level of financial risk within a firm, portfolio or position over a specific time frame. This metric is most commonly used by investment and insurance to determine the extent of loss that will not be exceeded for a given level of probability of occurrence

Risk managers use VaR to measure and control the level of risk exposure. One can apply VaR calculations to specific positions or whole portfolios or to measure firm-wide risk exposure.

99.5 percentile is VaR in 1-in-200 scenario

[2]

- ii) Spread risk is change in the value of portfolio due to change in the spread over the risk free rate of the bond. This rate is used to determine the bond price hence higher spread would mean lower price

Changes in the rating of the bond is referred to transition and risk of default or non – payment is referred as default risk. The transition of bond to the lower rating would result in higher spread and higher defaults.

The rating agencies provide the rating information which may be at lag compared to changes in the spread.

[3]

- iii) Spread risk.

- Get historical spread data for various ratings
- Fit the distribution
- Run simulations based on the distribution
- Estimate 1-in-200 spreads and corresponding change in value of portfolio

[3]

- iv) Transition and default risk

- Get historical transition and default data for various ratings
- Fit the distribution
- Run simulations based on the distribution
- Estimate 1-in-200 transition and default
- Estimate the default amount value of portfolio
- Estimate change in the market value of the bonds based on the transition.

[3]

- v) Approach 1

You can combine the 2 approaches by using the stressed spread when determining the movement of the portfolio in transition risk calculation

Approach 2

You can change the rating of the bonds and apply higher stress based on the transition & Default model.

[3]

- vi) VaR assesses the loss at a specific confidence level whereas Tail VaR measures the average of losses in the event that the specified confidence level is breached. So T-VaR more accurately represents extreme events while VaR ignores losses beyond the defined level of occurrence probability. When the fitted distributions are fat tailed T-VaR provides more accurate information. There may be situations when two portfolios when merged end up with a VaR more than the sum

of VaR of two portfolios which is not a desirable feature of any risk metric. T-Var doesn't have such an inconsistency. [2]

[16 Marks]

Solution 2:

- i) A yield curve is a line that plots the interest rates, at a set point in time, of bonds having equal credit quality but differing maturity dates. A yield curve is typically upward sloping; as the time to maturity increases, so does the associated interest rate. The reason for that is that debt issued for a longer term generally carries greater risk because of the greater likelihood of inflation or default in the long run. [2]
- ii) The inverted yield curve is an interest rate environment in which long-term debt instruments have a lower yield than short-term debt instruments of the same credit quality. This type of yield curve is the rarest of the three main curve types and is considered to be a predictor of economic recession.
A partial inversion occurs when only some of the short-term Treasuries (five or 10 years) have higher yields than 30-year Treasuries. An inverted yield curve is sometimes referred to as a negative yield curve. [2]
- iii) When investors are willing to accept lower returns for longer term bonds it indicates that they may not be too worried about inflation risk which often happens when the investors expect economy to not do too well in the future. In the past inverted yield curve has reflected investor's fear of recession where they decide to invest more on the long term bonds expecting future decline in the growth potential. The situation may also arise when they expect the central bank to keep cutting the short term interest rates (maybe to shore up the economy) and hence expect the long term interest rates to slip below the levels at which they currently are. [2]
- iv) Forward rate can become negative when yield curve inversion happens.

	Spot Rate(in %)		Forward rates
1	5	0.952381	
2	6	0.889996	7.01%
3	7	0.816298	9.03%
4	5	0.822702	-0.78%
5	4	0.821927	0.09%
6	4	0.790315	4.00%
7	3.5	0.785991	0.55%
8	3.5	0.759412	3.50%

[3]

[9 Marks]

Solution 3:

- i) Two different approaches are:
a. Implied volatility
b. And historical information [2]
- ii) Implied volatility is preferred approach for pricing if it is tradable instrument else use historical volatility.
Key limitation: implied volatility would be impacted by the liquidity in the market

Historical data method maybe used when adequate past data is available and conditions that existed in the past is a good proxy to what is to be expected in the future.

Key Limitations: Historical estimation may have biases of past data which may no longer be applicable for future. It also does not allow for conditional view of the market for example fall in market may result in higher volatility.

[5]

[7 Marks]**Solution 4:**

Use currency forward contracts for hedging the exposure. A currency forward contract is a binding contract in the foreign exchange market that locks in the exchange rate for the purchase or sale of a currency on a future date

Allowing for the interest rate and no arbitrage principle, we get the following spot rates.

	Portfolio 1	Portfolio 1	Forward rate
Rate of return	7%	2%	
Current value	20	1	20
1	21.40	1.02	20.98
2	22.90	1.04	22.01
3	24.50	1.06	23.09

[5 Marks]**Solution 5:**

	Calls
fair value (V)	$Se^{-q\tau}\Phi(d_1) - e^{-r\tau}K\Phi(d_2)$
delta (Δ)	$e^{-q\tau}\Phi(d_1)$
vega ($\mathcal{V}$)	$Se^{-q\tau}\phi(d_1)\sqrt{\tau} = Ke^{-r\tau}\phi(d_2)\sqrt{\tau}$
theta (Θ)	$-e^{-q\tau}\frac{S\phi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2) + qSe^{-q\tau}\Phi(d_1)$
rho (ρ)	$K\tau e^{-r\tau}\Phi(d_2)$
lambda (λ)	$\Delta\frac{S}{V}$
gamma (Γ)	$e^{-q\tau}\frac{\phi(d_1)}{S\sigma\sqrt{\tau}} = Ke^{-r\tau}\frac{\phi(d_2)}{S^2\sigma\sqrt{\tau}}$

You can delta hedge the portfolio by **buying** 20,494 shares of underlying.

Week	Stock Price	d1	D = N(d1)	No. of shares to be purchased
0	\$47.50	-0.228	0.410	=N(d1)*50,000 =20,494

[7 Marks]

Solution 6:

			LIBOR yield	Discount
Strike Swap Rate (s_k)	6.50%	Time	Curve	Factor
Forward Swap Rate Continuous Compounding		4.0	6.00%	0.000000
Forward Swap Rate m -Annual Compounding (s_0)	0.0717	5.0	6.00%	0.740818
Volatility of Forward Swap Rate (σ)	20%	6.0	6.50%	0.677057
Notional Principal (L)	100	7.0	6.50%	0.634448
Number of Payments per year (m)	1			
Sum of Discount Factors (A)	2.0523			
Option Maturity (T)	4			
Payer Swaption (P)	P			
Swaption Price	2.98			

[6 Marks]

Solution 7:

- i) Rearrange the AR(1) process $X_t - b = a(X_{t-1} - b) + \varepsilon_t$ to yield $X_t - X_{t-1} = (1 - a)(b - X_{t-1}) + \varepsilon_t$, which is a discrete time analog version of the Vasicek model. Recall that the autocorrelation of X_t at lag one is a , which translates to $(1 - \beta dt)$ in the Vasicek model. [3]
- ii) The spot rate curve is $r(t, T) = (-\ln P(t, T)/(T - t)) = -\ln A(t, T)/(T - t) + B(t, T)r(t, T)/(T - t)$
So the spot rate volatility becomes [3]

$$\frac{\partial r(t, T)}{\partial r} \sigma(r, t) = \frac{\sigma(r, t) B(t, T)}{T - t}$$

- iii) Changes in slope are due to changes in a , changes in curvature are due to changes in b and parallel moves are due to changes in r . [3]
- iv) The normal model does, for the following reason: It has fatter left tails and thinner right tails for probability distributions of interest rates in the future. It therefore gives thinner left tails and fatter right tails for the probability distribution of bond prices. [2]
- v) If every node contains forward rates up to the maturity of the underlying bond, a term structure up to that maturity can be constructed. The tree then needs to extend to only the maturity of the claim [3]

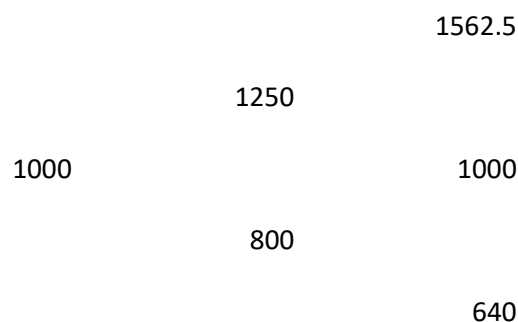
[14 Marks]

Solution 8:

- i) $1250p + 800(1-p) = 1000 \cdot \exp(0.05)$
 $p = 0.55838$ and therefore $(1-p) = 0.44162$ [2]

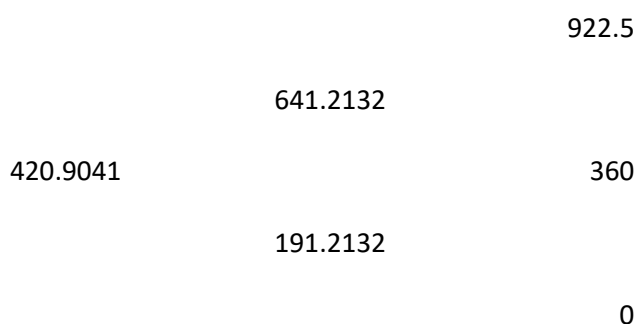
ii)

a. The stock price tree is



[2]

b. The option payoffs are:



Thus, the option price is 420.9041.

[3]

[7 Marks]**Solution 9:**

- i) A digital option is a type of options contract that has a fixed payout if the underlying asset moves past the predetermined threshold or strike price. In this case the asset is drift less so when the American pays off there is precisely a 50% chance (in the risk neutral measure) that the European will pay off too. So, the European is worth half as much as the American. [4]
- ii) A barrier option is a type of derivative where the payoff depends on whether or not the underlying asset has reached or exceeded a predetermined price. A barrier option can be a knock-out, meaning it expires worthless if the underlying exceeds a certain price, limiting profits for the holder and limiting losses for the writer. It can also be a knock-in, meaning it has no value until the underlying reaches a certain price. In this case the corresponding synthetic option will be a European Call Option [4]
- iii) A down – and – in call comes alive only when the barrier is touched; so, an increase in the stock price S brings it away from the barrier. This is not favorable. So, this is the correct answer (choice ii). For the other cases,
- A down – and – out call (option i) where the barrier has not been touched is still alive and hence benefits from an increase in S .
 - An up – and – in put (option iii) would benefit from an increase in the stock price as this would bring it closer to the barrier of \$110.
 - Finally, an up – and – in call (option iv) would also benefit if the stock price gets closer to the barrier. [4]

[12 Marks]

Solution 10:

- i) If the correlation is one, all names will default at the same time, and the junior and senior tranche will be equally affected. Hence, their spread should be 100 bps., which is the same for the collateral. With lower correlations, the losses will be absorbed first by the junior tranche. Therefore, the spread on the junior tranche should be higher, which is offset by a lower spread for the senior tranches. Hence J (which is the junior tranche spread) increases relative to S (the senior tranche spread). [3]
- ii) If Bank B buys company C, the two entities B and C shall default at the same time. This increase in the default correlation makes the CDS contract less valuable. The fair CDS spread decreases when the correlation increases. Given that the existing CDS contract has a fixed spread, this event should decrease the value of the outstanding contract, Hence the CDS value decreases. [3]
- iii) The answers are:
- Likely defaults are 14 $(1 - (1 - 0.015/0.5)^5) * 100$ [1.5]
 - Now, with a recovery rate of 50%. The expected loss is 7% of the notional. So all tranches up to the 7% point are wiped out. [1.5]
- iv) Outflow: $12\% - 3\% = 9\%$ and the inflow is $11\% + 0.4\% = 11.4\%$
- Thus, Net receipt: $11.4\% - 9\% = 2.4\%$
 - Notional principal: \$200 million
 - Thus, net receipt is for Bank One is: $\$200 \text{ million} * 2.4\% * 1 \text{ year} = \4.8 million . [3]

[12 Marks]**Solution 11:**

Please note that the PV of coupons is calculated as $[25 \exp(-0.088 * 0.25) + 25 \exp(-0.094 * 0.75)]$ and the bond future price is calculated as equal to $[(473.88 - 47.7542) \exp(0.098 * (11.5/12))]$.

Forward Price	468.0856
Strike Price	495
Risk-Free Rate	9.80%
Volatility of Forward Price	8.00%
Time to Maturity (years)	0.958333
Price of Call	4.787217

[5 Marks]
